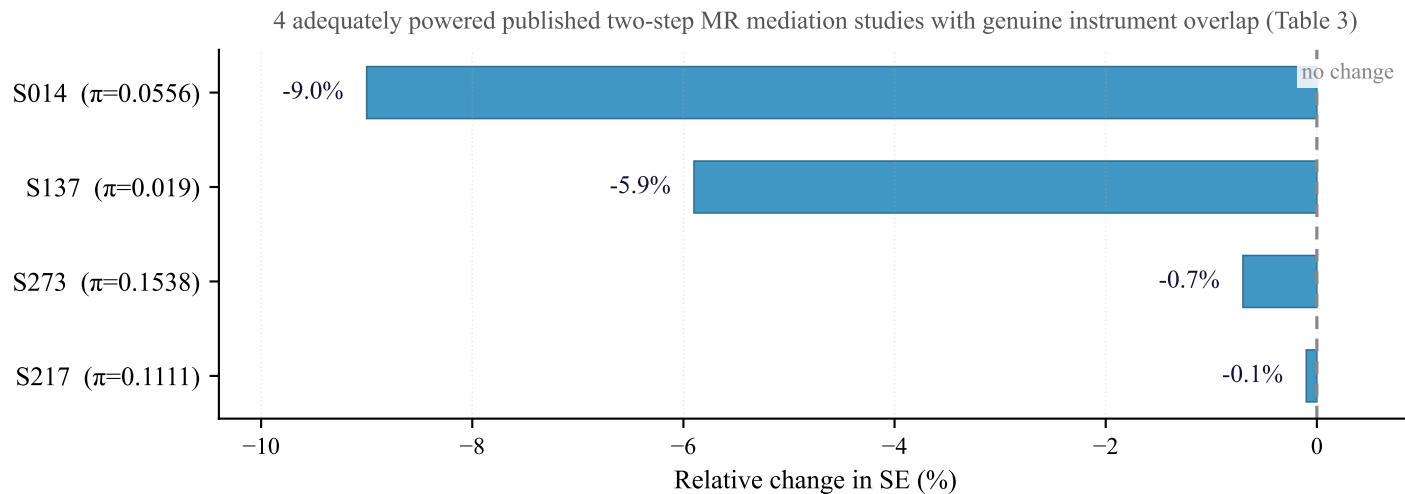


Supplementary Figure S4. Change in indirect-effect SE after S10 correction (TwoSampleMR-validated batch re-estimation)



TwoSampleMR-calibrated (cross-validated against the deposited D56 ground truth); all values negative = CI narrows; zero-flip conclusion fully validated.